

TOLEDO EXPRESS, OH, USA

WMO: 725360

Lat: 41.587N

Lon: 83.806W

Elev: 674

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94830

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.4	5.8	-7.0	3.9	1.3	-1.7	5.1	7.7	28.4	25.4	25.9	22.3	8.4	240	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	91.2	74.1	88.5	72.4	85.8	70.9	76.9	86.5	75.3	84.0	73.6	81.9	10.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	130.3	82.1	72.5	123.2	80.1	71.0	117.1	78.3	40.7	86.3	39.1	84.3	37.6	82.2	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.5	20.8	18.6	-4.9	95.4	6.7	3.1	-9.8	97.7	-13.7	99.5	-17.5	101.3	-22.3	103.6
			-5.4	79.7	6.3	1.9	-10.0	81.1	-13.7	82.2	-17.3	83.3	-21.9	84.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	25.5	28.3	37.5	49.5	60.5	70.2	73.6	71.8	65.2	53.4	41.4	31.1
	DBStd	19.13	11.88	11.03	10.73	9.16	8.45	6.69	5.57	5.26	7.36	8.64	9.20	10.11
	HDD50	2852	761	612	412	117	10	0	0	0	1	61	288	590
	HDD65	6028	1224	1029	853	473	193	27	3	8	86	373	709	1050
	CDD50	3136	2	3	25	101	335	605	731	676	458	166	29	5
	CDD65	838	0	0	1	7	54	181	269	219	93	14	0	0
	CDH74	7554	0	0	12	90	573	1667	2535	1746	805	124	2	0
	CDH80	2520	0	0	1	15	169	608	938	518	246	25	0	0
Wind	WSAvg	8.3	10.3	9.9	9.6	10.2	8.4	7.2	6.5	5.9	6.2	7.9	8.9	9.3
Precipitation	PrecAvg	34.3	2.2	2.2	2.4	3.3	3.6	3.5	3.3	3.2	2.9	2.6	2.5	2.4
	PrecMax	46.1	4.1	5.4	5.1	6.5	6.4	6.1	8.1	8.8	5.5	6.5	7.1	5.3
	PrecMin	27.8	0.7	0.5	0.7	1.0	1.3	0.9	1.0	0.7	1.0	0.3	0.9	0.6
	PrecStd	4.9	0.9	1.3	1.1	1.3	1.7	1.4	1.6	1.8	1.4	1.4	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	62.9	75.0	82.5	89.0	94.1	95.3	92.2	91.2	83.6	70.6	61.9
		MCWB	54.8	53.8	61.7	64.6	69.3	74.0	76.5	76.0	73.5	69.2	57.4	57.2
	2%	DB	53.1	55.0	67.0	76.0	84.7	90.2	91.4	88.5	86.5	77.2	64.5	55.5
		MCWB	50.3	50.5	56.6	60.2	67.4	72.3	74.8	73.8	70.3	64.0	56.8	51.6
	5%	DB	47.3	48.6	60.6	71.3	80.7	86.8	88.4	85.4	82.4	72.6	60.5	50.1
		MCWB	43.9	43.5	52.5	57.3	66.2	70.6	73.1	72.0	68.7	62.1	54.2	45.8
	10%	DB	41.0	42.4	54.4	65.9	76.1	83.2	85.3	82.6	78.6	68.1	56.0	44.5
		MCWB	38.1	38.3	47.8	54.5	64.0	69.3	71.4	70.5	66.7	60.1	50.2	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.9	57.2	62.7	67.1	73.5	77.0	79.8	78.8	76.4	71.4	62.1	58.1
		MCDB	58.0	61.0	73.6	78.8	83.4	88.5	90.7	89.2	86.7	79.9	66.6	61.3
	2%	WB	50.9	50.5	57.9	62.9	70.4	74.8	77.2	76.6	73.4	67.5	58.7	52.5
		MCDB	52.5	54.3	64.5	72.7	80.4	86.0	87.3	84.5	81.3	73.9	63.1	55.1
	5%	WB	44.3	43.8	53.8	59.3	68.2	73.2	75.4	74.8	71.2	64.3	54.8	46.2
		MCDB	46.6	48.2	59.8	68.2	77.1	83.2	84.4	81.4	78.2	70.2	59.6	49.2
	10%	WB	38.1	38.8	48.2	56.0	65.7	71.4	73.7	72.9	69.0	61.0	50.8	41.5
		MCDB	40.6	42.2	54.3	64.5	73.8	79.6	81.9	79.6	75.6	67.1	55.6	44.6
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.5	17.2	19.9	20.5	20.2	20.4	19.3	20.9	19.1	15.4	12.2
		MCDBR	18.1	21.2	26.2	27.5	26.1	24.4	22.7	21.8	25.2	25.3	21.5	17.3
		MCWBR	16.0	17.2	17.5	15.6	12.4	10.2	9.5	9.8	11.6	13.7	15.4	14.7
	5% WB	MCDBR	17.4	20.3	24.0	23.9	22.7	20.9	19.9	18.1	20.5	21.2	19.3	16.2
		MCWBR	16.2	17.6	17.7	15.7	12.4	10.3	9.5	9.1	10.7	13.1	15.8	15.0
Clear-Sky Solar Irradiance	taub	0.298	0.334	0.358	0.404	0.433	0.450	0.442	0.434	0.387	0.359	0.324	0.309	
	taud	2.408	2.294	2.317	2.208	2.180	2.176	2.214	2.223	2.338	2.397	2.453	2.404	
	Ebn at Noon	271	275	281	274	268	263	264	262	268	263	259	255	
	Edh at Noon	25	33	36	44	46	46	44	43	36	30	24	23	
All-Sky Solar Radiation	RadAvg	489	779	1109	1468	1717	1917	1927	1687	1375	908	580	403	
	RadStd	52	105	100	126	151	129	106	66	111	75	79	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	N/A

Nomenclature: See separate page